

COMP1511

Introduction to Discrete Mathematics

Lecture Notes

2023/24

The three main topics that are covered in this module are:

1. Combinatorics: Counting Methods
2. Discrete Probability Theory
3. Graph Theory

The lecture notes closely follow the material presented in Chapters 6, 7, 10 and 11 of the accompanying book (that is referred to in the rest of the text as *the book*):

K. H. Rosen, **Discrete Mathematics and its Applications**, 7th Edition, McGraw Hill Education, 2013.

All of the material from the module syllabus will be covered in the lecture notes and presented in the lectures. The examination and coursework will only assess topics covered in the module syllabus. You should use the book as an additional source of problems to solve and as an alternative explanation of a topic. The book is there to supplement the lecture notes. Not all of the material from the chapters is covered in the lecture notes (i.e. lectures), and the lectures sometimes cover the material in a different order than the one in the book.

We expect you, as a student, to:

1. Attend lectures and carefully read the lecture notes.
2. Attend and engage in tutorials, completing all problems requested by your tutorial tutor.
3. Submit all courseworks.
4. To truly absorb the material covered, you should also practice exercises from the relevant sections of the book (the odd numbered ones have solutions in the back).

1 Combinatorics: Counting Methods **(Corresponds to Sections 6.1 - 6.5 in the book)**

2 Discrete Probability Theory **(Corresponds to Sections 7.1 - 7.3 in the book)**

An *experiment* is a process that yields an outcome.

The set S of all possible outcomes of a given experiment is called the *sample space*.

An *event* A is a set of outcomes, i.e. a subset of the sample space S .

Example

Experiment: rolling a six-sided die

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$

Event: obtaining a 4 when rolling a die, i.e. $A = \{4\}$

Since an event is a set, we can combine events to form new events using the various set operations:

- (i) $A \cup B$ is the event that occurs if and only if A occurs *or* B occurs (or both).
- (ii) $A \cap B$ is the event that occurs if and only if A occurs *and* B occurs.
- (iii) $\bar{A}$, the complement of A , is the event that occurs if and only if A does *not* occur.

Two events A and B are called *mutually exclusive* if they are disjoint, i.e. $A \cap B = \emptyset$ (they cannot occur simultaneously). Three or more events are mutually exclusive if every two of them are mutually exclusive.

Example

Let A be the event that an even number occurs, B that an odd number occurs, C that a prime number occurs.

$$A = \{2, 4, 6\}, B = \{1, 3, 5\}, C = \{2, 3, 5\}$$

$A \cup C = \{2, 3, 4, 5, 6\}$ the event that an even or a prime number occurs

$B \cap C = \{3, 5\}$ the event that an odd prime number occurs

$\bar{C} = \{1, 4, 6\}$ the event that a prime number does not occur

A and B are mutually exclusive.

Example

Experiment: toss a coin 3 times and observe the sequence of heads (H) and tails (T).

$S =$

$A =$ event that 2 or more heads appear consecutively $=$

$B =$ event that five heads appear $=$

2.1 Finite Probability Spaces

Let S be a finite nonempty sample space, say $S = \{a_1, a_2, \dots, a_n\}$.

A *finite probability space* or *probability model*, is obtained by assigning to each element a_i in S a real number p_i , called the *probability* of a_i , satisfying:

- (i) each p_i is nonnegative, i.e. $p_i \geq 0$, and
- (ii) the sum of p_i 's is 1, i.e. $p_1 + p_2 + \dots + p_n = 1$.

The *probability* of an event A , written $P(A)$, is the sum of the probabilities of the elements in A .

The singleton set $\{a_i\}$ is called an *elementary event*, and for notational convenience we write $P(a_i)$ for $P(\{a_i\})$.

Example

Experiment: a loaded die is tossed.

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$.

The following assignment defines a probability space: $P(1) = \frac{3}{8}$, $P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{8}$.

Let A be the event that an odd number appears, and B the event that an even number appears.

$$P(A) =$$

$$P(B) =$$

Example

Experiment: let 3 coins be tossed and the number of heads observed.

Sample space: $S = \{0, 1, 2, 3\}$.

The following assignment defines a probability space: $P(0) = \frac{1}{8}$, $P(1) = \frac{3}{8}$, $P(2) = \frac{3}{8}$, $P(3) = \frac{1}{8}$.

Let A be the event that at least one head appears, and let B be the event that all heads or all tails appear.

$$P(A) =$$

$$P(B) =$$

2.2 Equiprobable Spaces

A finite probability space S where each element of S has the same probability is called an *equiprobable space*.

In particular, if S contains n elements then the probability of each element of S is $\frac{1}{n}$.

For an event A of an equiprobable space S , $P(A) = \frac{\text{number of elements in } A}{\text{number of elements in } S} = \frac{|A|}{|S|}$.

Example

Let a fair die be tossed.

A = event that an odd number appears

$$P(A) =$$

Example

Let a card be selected from an ordinary deck of 52 cards.

A = event that the card is a spade

B = event that the card is a face card (jack, queen or king)

$$P(A) =$$

$$P(B) =$$

$$P(A \cap B) =$$

Example

In poker player gets 5 cards. Number of different poker hands is $C(52, 5)$.

(a) Full house: 3 cards of one rank, 2 cards of another rank.

What is the probability of receiving a full house?

(b) What is the probability of receiving a four of a kind?

Theorem 2.1 *Let A be an event. Then $P(A) + P(\bar{A}) = 1$.*

Note: this theorem is useful when it is easier to compute $P(\bar{A})$ than $P(A)$.

Example

Five distinct fair dice are rolled. What is the probability of getting at least one 6?

Example

Find the probability that among n people, at least two people have birthdays on the same month and date (but not necessarily in the same year). Assume that all months and dates are equally likely, and ignore February 29 birthdays.

Theorem 2.2 *Let A_1 and A_2 be events. Then $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$.*

Example

A fair coin is tossed 3 times, and a sequence of heads and tails is observed. What is the probability that an even number of heads or at least two tails appears?

Example

Two different colored fair dice are rolled. The sample space S consists of ordered pairs (a,b) where a and b are integers from 1 to 6. What is the probability of getting doubles (two dice showing the same number) or a sum of 6?

Corollary 2.1 *If A_1 and A_2 are mutually exclusive events, then $P(A_1 \cup A_2) = P(A_1) + P(A_2)$.*

2.3 Conditional Probability

Example

Suppose that we roll two different colored fair dice. What is the probability of getting a sum of 10? What if we are told at least one die is a 5?

$$|S| = 36$$

(1, 1)	(2, 1)	(3, 1)	(4, 1)	(5, 1)	(6, 1)
(1, 2)	(2, 2)	(3, 2)	(4, 2)	(5, 2)	(6, 2)
(1, 3)	(2, 3)	(3, 3)	(4, 3)	(5, 3)	(6, 3)
(1, 4)	(2, 4)	(3, 4)	(4, 4)	(5, 4)	(6, 4)
(1, 5)	(2, 5)	(3, 5)	(4, 5)	(5, 5)	(6, 5)
(1, 6)	(2, 6)	(3, 6)	(4, 6)	(5, 6)	(6, 6)

Probability of getting a sum of 10 = $\frac{3}{36} = \frac{1}{12}$ (ovals).

If we are told that at least one die is 5 (cross), the probability of getting a sum of 10 = $\frac{1}{11}$ (cross and ovals)

A probability given that some event occurred is called *conditional probability*. Suppose E is an event in a sample space S with $P(E) > 0$. The probability that event A occurs once E has occurred, the *conditional probability of A given E* , written $P(A|E)$ is defined by:

$$P(A|E) = \frac{P(A \cap E)}{P(E)}.$$

Theorem 2.3 Suppose S is an equiprobable space and A and E are events. Then

$$P(A|E) = \frac{|A \cap E|}{|E|}.$$

Example

A fair coin is tossed 10 times, giving heads on each toss. What is the probability of a toss of heads on the eleventh toss?

Example

A pair of different colored fair dice is tossed. Find the probability that one of the dice is 2 if the sum is 6.

2.4 Independent Events

Events A and B in probability space S are *independent* if the occurrence of one of them does not influence the occurrence of the other.

In other words, B is independent of A if $P(B) = P(B|A)$.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow P(A \cap B) = P(A)P(B|A) = P(A)P(B).$$

So we can use the following definition of independence:

Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$; otherwise they are *dependent*.

Example

A fair coin is tossed 3 times. Let A be the event that there is at least one head, and B the event that there is at least one tail. Are events A and B independent?

Example

Intuitively, if we flip a fair coin twice, the outcome of the second toss does not depend on the outcome of the first toss.

E_H = event that head is on first toss

E_T = event that tail is on second toss.

Prove that E_H and E_T are independent.

Example

Joe and Alice take a final exam in math. The probability that Joe passes is 0.70 and the probability that Alice passes is 0.95. Assume that events J = “Joe passes”, and A = “Alice passes” are independent. Find the probability that Joe or Alice (or both) pass the final exam.

2.5 Bayes' Theorem

We can find the conditional probability that an event F occurs, given that an event E has occurred, when we know $P(E|F)$, $P(E|\bar{F})$ and $P(F)$.

Theorem 2.4 (Bayes' Theorem) *Suppose that E and F are events from a sample space S such that $P(E) > 0$ and $P(F) > 0$. Then*

$$P(F|E) = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|\bar{F})P(\bar{F})}.$$

Example

Suppose that one person in 100,000 has a particular rare disease for which there is a fairly accurate diagnostic test. This test is correct 99% of the time when given to a person selected at random who has the disease; it is correct 99.5% of the time when given to a person selected at random who does not have the disease. Given this information can we find

- (a) the probability that a person who tests positive for the disease has the disease?
- (b) the probability that a person who tests negative for the disease does not have the disease?

Should a person who tests positive be very concerned that he or she has the disease?

F = event that a person selected at random has a disease

E = event that a person selected at random tests positive for the disease

- (a) We want $P(F|E)$ (the probability that a person who tests positive for the disease actually has the disease)